



Computer Engineering Department
Computer Engineering
IC6400 - Operations Research

Transport Problems with Excel / Calc Solver

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What is Microsoft Excel Solver and How to use it?

Solver is a Microsoft Excel add-in program used for what-if analysis. Solver is used to find an optimal (maximum or minimum) value for a formula in one cell, called the objective cell, subject to constraints, or limits, on the values of other formula cells on a worksheet. Solver works with a group of cells, called decision variables or simply variable cells, that are used in computing the formulas in the objective and constraint cells. Solver adjusts the values in the decision variable cells to satisfy the limits on constraint cells and produce the result for the objective cell.

To add the Solver program in Microsoft Excel, go to the File section in the upper part of the Excel menu. Inside it, find the Options button click it, this will open the Options window. Go to the bottom and click the Go button, this will open the Add-Ins window. Find the Solver Add-in and check the box next to it, then click OK. After this, the Solver program will be available in the Data tab.

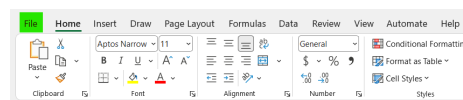


Figure 1: File section

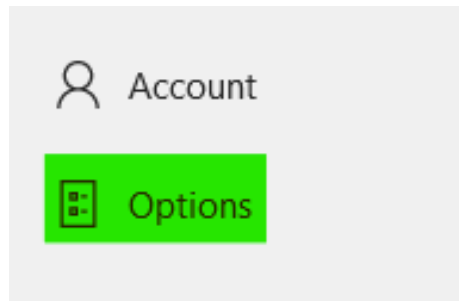


Figure 2: Options button

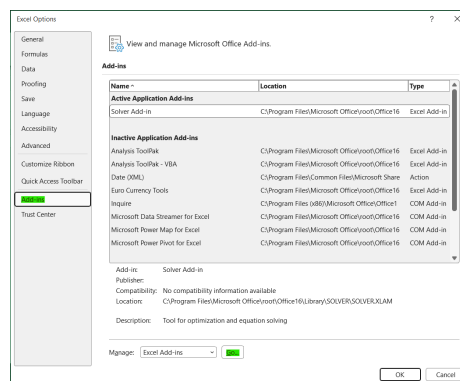


Figure 3: Excel Options

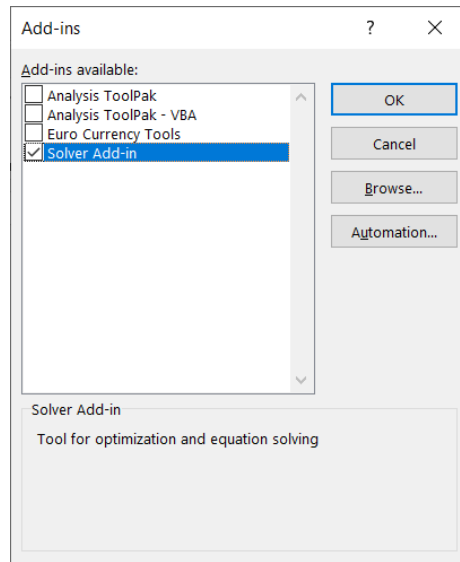


Figure 4: Add-Ins

To use Solver, go to the Data tab, in the Analysis group and select Solver. A window will pop to set the problem parameters, in the Set Objective box, you can enter a cell reference or name for the objective cell. The objective cell must contain a formula.

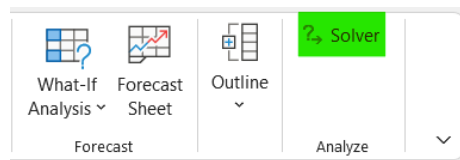


Figure 5: Solver Button

After setting the objective cell, select the option "Max" to maximize the formula, "Min" to get the minimum value, use "Value of" to set a certain value for the objective cell. In the By Changing Variable Cells box, enter a name or reference for each decision variable cell range. Separate the nonadjacent references with commas. The variable cells must be related directly or indirectly to the objective cell.

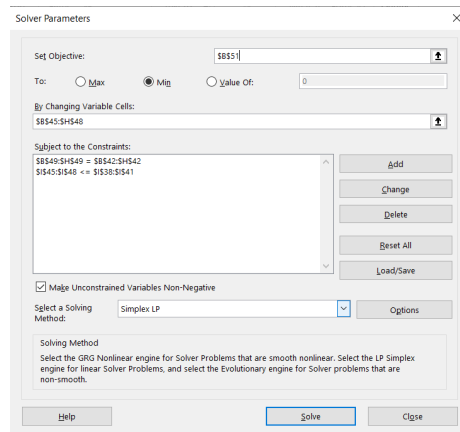


Figure 6: Excel Solver

In the Subject to the Constraints box will appear all the constrain added to the problem. To add a constrain, select Add, enter the cell reference or name of the cell range in the "Cell Reference" box and select the relationship between the referenced cell and the constraint.

It is possible to change how Solver find solutions by choosing a solving method. You can choose between "Generalized Reduced Gradient (GRG) Nonlinear", "LP Simplex" and "Evolutionary".

After setting all constraints and choosing how to solve the problem, select Solve. You can keep the solution values on worksheet by selecting "Keep Solver Solution" in the "Solver Results" dialog box, or choose restore the original values by selecting "Resotore Original Values".

How to use Libre Office Calc Solver?

LibreOffice Calc has also a built-in Solver. To access Solver choose "Tools" on the manu bar and then select Solver. It will also pop a window to set the problem parameters. In the "Target Cell" box you can enter the target cell reference.

Then you can select "Maximum" to optimize the problem to find the maximum value of the taget cell or select "Minimum" to minimize the target cell, or select "Value of" to approach the target cell to a given value.

Enter the cell range that can be changed in the "By Changing Cells" box, these are the variables of the equiation. To add a set of constraints for the problem you can use the "Limiting Conditions" section, each constraint is represented by a cell reference, an operator and a value.

After everyting is set, click "Solve" to solve the problem. The dialog settings are retained until you close the current document.

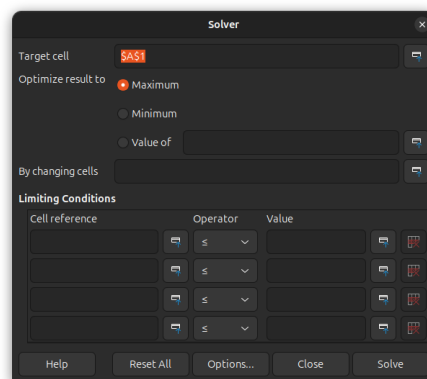


Figure 7: Calc Solver

Setup

In the next two chapters, there will be images with three distinct parts for the solver to process:

Table of costs of production

This table contains the cost for production from some producer i to some consumer j . The maximum amount of production each producer can generate, which correspond to the numbers located on the right side of the table and the needed amount of consumption for each consumer located on the bottom of the table.

Changing cells

This table has cells that are given to the solver to make the calculations and find the optimal solution. This table needs some setup before it is used. For each row, calculate the sum of each cell and put the result on the right of the row and also calculate the sum for the cells in each of the columns and put the result on the bottom of the column. At the end, the solver puts the values needed in every cell to construct the optimal solution.

Total cost

This cell will contain the optimal solution when the solver has done all the calculations. The cell needs some setup before it is used. In the cell calculate the sum product of the table of costs and the table of changing cells. The name of this function is `SUMPRODUCT` and it can receive multiple matrices as inputs, as long as every matrix has the same dimensions.

Transport Problems

Plantas Hidroeléctricas					
	C1	C2	C3	C4	
P1	8	6	10	9	35
P2	9	12	13	7	50
P3	14	9	16	5	40
	45	20	30	30	
	C1	C2	C3	C4	
P1	0	10	25	0	35
P2	45	0	5	0	50
P3	0	10	0	30	40
	45	20	30	30	
Total Cost	1020				

Figure 8: Hydroelectric Plants

Ejemplo 2				
	C1	C2	C3	
P1	7	8	10	50
P2	9	7	8	50
P3	10	9	13	20
	40	40	40	
	C1	C2	C3	
P1	40	10	0	50
P2	0	10	40	50
P3	0	20	0	20
	40	40	40	
Total Cost	930			

Figure 9: Example 2

Plantas Hidroeléctricas (modificado)						
	C1	C2	C3	C4	Dummy	
P1	8	6	10	9	0	35
P2	9	12	13	7	0	50
P3	14	9	16	5	0	40
	40	20	30	30	5	
	C1	C2	C3	C4	Dummy	
P1	0	15	20	0	0	35
P2	40	0	10	0	0	50
P3	0	5	0	30	5	40
	40	20	30	30	5	
Total Cost	975					

Figure 10: Modified Hydroelectric Plants

Plantas Hidroeléctricas (modificado 2)				
	C1	C2	C3	
P1	8	6	9	50
P2	14	9	5	50
Dummy	20	22	23	20
	40	40	40	
	C1	C2	C3	
P1	20	30	0	50
P2	0	10	40	50
Dummy	20	0	0	20
	40	40	40	
Total Cost	1030			

Figure 11: Modified Hydroelectric Plants 2

Problema de Transporte Degenerado						
	C1	C2	C3	C4	Dummy	
P1	10	2	20	11	0	15
P2	12	7	9	20	0	25
P3	4	14	16	18	0	10
	5	15	10	15	5	
	C1	C2	C3	C4	Dummy	
P1	0	0	0	15	0	15
P2	0	15	10	0	0	25
P3	5	0	0	0	5	10
	5	15	10	15	5	
Total Cost	380					

Figure 12: Degenerated Transport Problems

Transferring Problems

Fábrica de Computadores						
	NY	Chicago	LA	Boston	Dummy	
Memphis	8	13	25	28	0	150
Denver	15	12	16	25	0	200
NY	0	6	16	17	0	350
Chicago	6	0	14	16	0	350
	350	350	130	130	70	
	NY	Chicago	LA	Boston	Dummy	
Memphis	60	0	0	0	70	130
Denver	0	0	130	70	0	200
NY	290	0	0	60	0	350
Chicago	0	350	0	0	0	350
	350	350	130	130	70	
Total Cost	5330					

Figure 13: Computer Factory

Fábrica de Computadores (variante)								
	NY	Chicago	LA	Boston	Dummy	Memphis	Denver	
Memphis	8	13	25	28	0	0	7	500
Denver	15	12	16	25	0	11	0	550
NY	0	6	16	17	0	0	0	350
Chicago	6	0	14	16	0	0	0	350
	350	350	130	130	90	350	350	
	NY	Chicago	LA	Boston	Dummy	Dummy	Dummy	
Memphis	60	0	0	0	90	350	0	500
Denver	0	0	130	70	0	0	350	550
NY	290	0	0	60	0	0	0	350
Chicago	0	350	0	0	0	0	0	350
	350	350	130	130	90	350	350	
Total Cost	5330							

Figure 14: Computer Factory Variant

Fábrica de Computadores (variante 2)								
	NY	Chicago	LA	Boston	Dummy	Memphis	Denver	
Memphis	8	13	25	28	0	0	7	500
Denver	15	12	26	25	0	11	0	550
NY	0	6	16	17	0	0	0	350
Chicago	6	0	14	16	0	0	0	350
LA	0	0	0	11	0	0	0	350
Boston	0	0	19	0	0	0	0	350
	350	350	480	480	90	350	350	
	NY	Chicago	LA	Boston	Dummy	Memphis	Denver	
Memphis	150	0	0	0	0	350	0	500
Denver	0	0	0	110	90	0	350	550
NY	200	0	130	20	0	0	0	350
Chicago	0	350	0	0	0	0	0	350
LA	0	0	350	0	0	0	0	350
Boston	0	0	0	350	0	0	0	350
	350	350	480	480	90	350	350	
Total Cost	6370							

Figure 15: Computer Factory Variant 2